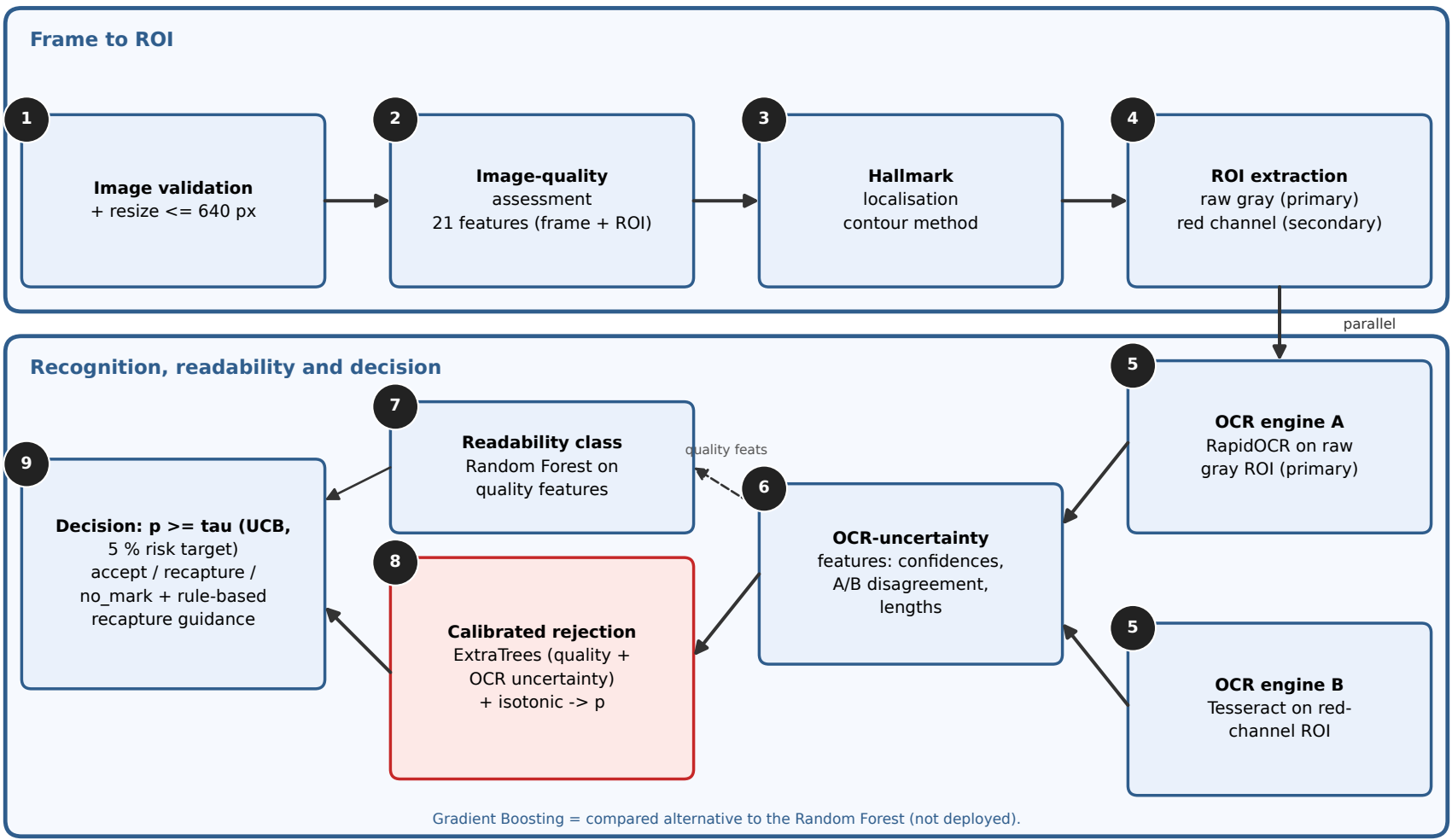
Companion processing pipeline (implemented, Python)
Frame to ROI
1
Image validation
+ resize <= 640 px
2
Image-quality
assessment
21 features (frame + ROI)
3
Hallmark
localisation
contour method
4
ROI extraction
raw gray (primary)
red channel (secondary)
parallel
Recognition, readability and decision
5
OCR engine A
RapidOCR on raw
gray ROI (primary)
7
Readability class
Random Forest on
quality features
quality feats
9
Decision: p >= tau (UCB,
5 % risk target)
accept / recapture /
no_mark + rule-based
recapture guidance
6
OCR-uncertainty
features: confidences,
A/B disagreement,
lengths
8
Calibrated rejection
ExtraTrees (quality +
OCR uncertainty)
+ isotonic -> p
5
OCR engine B
Tesseract on red-
channel ROI
Gradient Boosting = compared alternative to the Random Forest (not deployed).